Proof

Questions 1–14

1.45 — Question 1. In each case, choose one of the statements

$$A \Rightarrow B \qquad A \Leftarrow B \qquad A \Leftrightarrow B$$

to describe the relationship between A and B

a) A: The shape has 4 equal sides.
   B: The shape is a square.
b) A: A quadrilateral has perpendicular diagonals.
   B: The quadrilateral is a kite.
c) A: A quadrilateral has one pair of parallel sides.
   B: The quadrilateral is a trapezium.
d) A: $PQRS$ is a rhombus.
   B: $PQRS$ is a parallelogram.

1.45 — Question 2. In each case, choose one of the statements

$$A \Rightarrow B \qquad A \Leftarrow B \qquad A \Leftrightarrow B$$

to describe the relationship between A and B

a) A: $x = 3$ B: $(x-3)(x+2) = 0$
b) A: $x > 6$ B: $x = 10$
c) A: $x = 4$ B: $x^2 = 16$
d) A: $x^3 = -1$ B: $x = -1$

1.45 — Question 3. For each of the following determine which of these statements are true, where it is false explain your answer:

a) If n is an even integer, n^2 is even
b) The product of 5 consecutive integers is a multiple of 5
c) $x^2 > 0$ for all $x \in \mathbb{R}$
d) The square root of a positive number is always positive
e) The product of any two odd integers is always even
f) If $\cos x^\circ = 0.5$, then $x = 60$

1.45 — Question 4. Find the mistake in the proof below:

$$2 = \frac{20}{10} = \frac{10+10}{10} = \frac{(10-10)(10+10)}{10(10-10)}$$

$$= \frac{10^2 - 10^2}{100-100} = \frac{100-100}{100-100} = 1$$

1.45 — Question 5. For each statement, find an example that disproves it:

a) All prime numbers are odd.
b) All squares are even numbers.
c) All multiples of 3 are also multiples of 6
d) All real numbers have a square root.
e) All perfect squares end in an odd number.

1.45 — Question 6. For each statement, find an example that disproves it:

a) If $x^4 > x$, then $x > 1$
b) If n is a prime number, then $n^2 - n + 13$ is prime.
c) The value of $n^2 + n + 17$ is a prime number for all positive integer values of n.

1.45 — Question 7. Use proof by exhaustion to prove that

a) the sum of the digits of any 2-digit number is less than or equal to 18

b) if a 2-digit number is divisible by 3, then the sum of its digits is divisible by 3

c) 89 is a prime number.

1.45 — Question 8. Prove that no square number ends in a 3

1.45 — Question 9. Show that

a) the sum of 2 consecutive numbers is always odd.
b) the sum of any two consecutive odd numbers is a multiple of 4
c) the square of an odd number is always odd.
d) the sum of the squares of 2 consecutive odd numbers is always even.

1.45 — Question 10. Prove

a) the product of any three consecutive positive integers is a multiple of 6
b) the product of any four consecutive positive integers is a multiple of 24

1.45 — Question 11. Prove that for any even number n, the expression $6n^2 + 10n$ has a factor of 4

1.45 — Question 12. Prove that $n^4 - n^2$ is a multiple of 6 for all positive integer values of n

1.45 — Question 13. Prove that $n^3 - n$ is a multiple of 3 for all positive integer values of n, using

a) Proof by deduction

b) Proof by cases, by considering $n = 3k$, $n = 3k + 1$ and $n = 3k + 2$

1.45 — Question 14. For a positive integer number n, prove that $n^4 - 4$ is not divisible by 8